

Feature Checklist Audit

Every claim below was checked directly against the running code — file and line cited where it matters. Three states: **DONE** (matches as specified), **PARTIAL** (the idea exists but differs from the exact spec, or the data exists but isn't surfaced), **MISSING** (no trace in the code).

TIER 1 Must-Have	TIER 2 Strong Additions	TIER 3 Nice-to-Have	TIER 4 Don't Do
11 / 15	7 / 12	1 / 6	7 / 7 clean

Summary

Biggest gap: the entire “Reliability, Crash Prevention & Caching” section (checklist's own Tier 1 item #15) is missing. No request timeouts, no retry/backoff, no circuit breaker, no fallback chain beyond a flat cache, no global exception handler. The checklist itself calls this “a P0 risk, not polish” — worth prioritizing before a live demo on venue wifi.

Direct conflict with a change made earlier in this session: the checklist's “Dashboard Simplification” section asks for exactly the 4 primary metric cards + collapsible “Detailed Analytics” pattern that was just removed from the Dashboard (at your request) and replaced with the “Explore the Platform” navigation grid. Those metrics still exist on the Thermal Simulation page — just not on the Dashboard anymore.

Genuine strength: scientific-integrity language matches the checklist's honesty rules almost exactly. No invented accuracy figure (no “±2–3°C”), no false “ASHRAE 140 validated” claim, no fabricated future date — the app consistently says “Not yet validated” / “Field validation required before deployment.”

Core Functionality — the 3 SIH 26051 Requirements

These are the checklist's own “Tier 1: Absolute Must-Haves.”

REQUIREMENT 1 — SHELTER INSIDE TEMPERATURE PREDICTION

DONE	Hourly prediction, min/max range, temperature curve graph Two-node RC model stepped hourly; engine.js:172–279, rendered in ui-2.js Thermal Simulation page
PARTIAL	“Real-time update when design parameters change” Recomputes on an explicit “Run Simulation” click, not live as you edit a field
DONE	Solar gain, wall/roof/floor losses, ventilation losses, thermal mass exchange All present as separate terms — engine.js:197–270

PARTIAL

Occupancy heat generation accounted for

Only a flat manual “Internal heat gain (W)” number is used (`engine.js:221`); the “Occupancy (persons)” count is stored (`store.js:22`) but never read anywhere in `engine.js` — occupant count doesn’t actually drive the physics

REQUIREMENT 2 — THERMAL ENERGY FROM SOLAR RADIATION

DONE

Daily + annual solar (NASA POWER), sunshine window, orientation impact on gain

`nasa-power.js`, orientation factor table in `ARCHITECTURE.md §3.5`

PARTIAL

Orientation-specific recommendation text (“south-facing better in winter”, etc.)

The Evaluator Summary’s “Why this design?” bullets are real, computed reasoning — not the canned per-orientation sentences the checklist describes, but arguably more credible since they’re derived from actual results

REQUIREMENT 3 — HEAT FLOW DETAILS

PARTIAL

Hourly W-by-W breakdown

The engine computes hourly internally, but the Thermal Simulation UI’s “Heat Flow Analysis” panel only displays daily kWh totals per component, not an hourly time series of each loss term

DONE

Daily totals per component, net energy balance

`ui-2.js` Thermal Simulation page — `solar/wall/roof/floor/opening/ventilation/mass`, net kWh/day

PARTIAL

“Stacked bar chart” visual + direction indicators

Rendered as separate horizontal bar rows (green=gain, red=loss), not one single stacked/segmented bar

Climate & Location Features

REAL CLIMATE DATA INTEGRATION

DONE

Open-Meteo: temperature, humidity, wind, cloud cover — hourly, live

`weather-api.js:66-81`

MISSING

Open-Meteo precipitation

Not in the fetched parameter list (`weather-api.js:67`)

DONE

NASA POWER: GHI, DNI, daily/annual totals, 7-day cache

`nasa-power.js:38-64`

MISSING

NASA POWER diffuse radiation component

Only GHI + DNI parameters requested, no diffuse term

DONE

Data source badges (“Real Data” / live)

`util.js:11-18 U.badge()`, used throughout

MULTI-LOCATION SUPPORT

DONE

10 reference Indian locations, region + coordinates + elevation shown

`data.js:13-23`

MISSING	Manual coordinate entry, interactive map picker, climate-zone classification label Dropdown-only selection from the fixed 10; no map/geocoding UI found
PARTIAL	Elevation display Shown, but from a hardcoded reference table, not a live elevation API/DEM lookup

CLIMATE SUMMARY CARDS

PARTIAL	Temperature profile card (diurnal swing shown; monthly breakdown graph not) NASA POWER already fetches monthlyGhi/monthlyTemp (nasa-power.js:61-62) but neither is read or charted anywhere in ui-1.js — the data exists, it's just not surfaced
MISSING	Solar card's “best months for heating” / monsoon impact / seasonal graph Only the single annual figure is shown
PARTIAL	Humidity & wind card Wind/RH/cloud shown as one compact tile; no “ventilation need” calculation or monsoon warning
MISSING	Dedicated “Design implications” auto-recommendation card on the Location page No such card found — closest equivalent is the Evaluator Summary's post-simulation reasoning, which lives on a different page after a run

Shelter Design Features

PARTIAL	Shape selector Rectangular, Square, Circular, Dome, Semi-circular, Custom (ui-1.js:316) — more variety than asked, but no L-shaped option
DONE	Dimension inputs + derived floor/wall/roof area, volume ui-1.js:321-327, 374-377
MISSING	Surface-area-to-volume ratio displayed Derivable from shown values but not computed/shown as its own metric
DONE	All 8 compass directions + custom azimuth ui-1.js:335
MISSING	Sun position indicator 2D preview shows an “N↑” compass mark only, no sun-path/position graphic
DONE	Window area/count/face, glazing type (single/double/triple/Low-E) with U-value + SHGC shown, door area, ACH ui-1.js:344-353
MISSING	Door material/insulation selection Only door area is configurable, no door material picker
PARTIAL	2D top-down visualization ui-1.js:237-260 — shows shape, orientation and one generic front-face window mark; not labeled with dimensions and doesn't reflect the actual configured window/door count or placement
MISSING	3D visualization Explicitly out of scope by design — README.md: “no 3D preview (2D top-down schematic only)”

Material Database & Selection

DONE	Full property set: k, density, specific heat, absorptivity, emissivity (+ reflectivity, bonus); U-value calculated from layers data.js:31-63, engine.js:85-98
DONE	Wall (7/7), insulation (7/7), glazing (4/4), thermal mass (6/6 + “none”) options all match the checklist by name data.js:31-69
PARTIAL	Roof options RCC, Metal, Insulated Metal, Composite present; Thatch/natural roofing not in the library
MISSING	Custom material input (user-defined properties) No “add material” UI found across the Materials or Designer pages
MISSING	Regional availability (Yes/No, local sourcing, lead time, transport multiplier, supplier info) Not modeled — only a single coarse sustainability: HIGH/MEDIUM/LOW tag exists per material
PARTIAL	Cost database Flat costPerM2/costPerKg per material exists and feeds the optimizer's cost score (engine.js:344-355); no installation-complexity factor, waste factor, or CPWD/state-PWD Schedule-of-Rates citation — figures are unsourced reference numbers
MISSING	Detailed sustainability database (embodied carbon, recyclability %, water use, health impact, end-of-life) Only the single HIGH/MEDIUM/LOW tag exists, no per-metric breakdown

Thermal Simulation Engine & Occupancy

DONE	Hourly transient RC network, ground coupling, ventilation/infiltration, thermal mass response engine.js:149-279, documented in ARCHITECTURE.md §3
PARTIAL	Separate interior/exterior surface temperature nodes; radiation heat transfer as its own term By design a simplified 2-node model (indoor air + thermal mass) — sky longwave radiation is explicitly folded into the sol-air term rather than modeled separately (disclosed in Settings' Assumptions list)
MISSING	Occupancy Heat Generation Model — activity-based wattage presets (Sleeping ~85W, Resting ~120W, Light ~180W, Moderate ~240W, Heavy ~360W per person) No such mapping exists. Note: this session added a “clothing/activity level” selector, but it only shifts the comfort temperature band — it does not compute occupant heat output in Watts
MISSING	Sensible/latent heat split + occupancy-linked ventilation coupling + moisture balance No humidity/moisture node exists anywhere in engine.js — only the two temperature nodes
MISSING	Occupancy schedule (varying by hour) Internal gains are constant-per-hour only, disclosed as such in the Assumptions list

Visualization & Analytics

Dashboard's 4 primary metric cards were removed this session (Thermal Comfort Score / Solar Gain / Heat Loss / Avg Indoor Temp), at your explicit request, and replaced with the “Explore the Platform” navigation grid. They still exist — just on the Thermal Simulation page now, not the Dashboard. The checklist's “Dashboard Simplification” section asks for the exact pattern that was just removed.

PARTIAL

Comfort score explicitly labeled as a custom index, not PMV/PPD

The formula is documented as a “weighted blend” in `ARCHITECTURE.md §3.8`, but the live UI itself doesn't carry an explicit “this is not a PMV/PPD standard” disclaimer next to the score

DONE

24h indoor-vs-ambient chart with comfort-band shading

`ui-2.js:225, charts.js`

MISSING

4-tier color-coded temperature curve (green/yellow/red/blue by zone)

Only a single green comfort band is shaded; the curve itself isn't segment-colored by temperature zone

MISSING

Sankey diagram, pie chart for heat flow

`charts.js` implements line/bar/gauge charts only

DONE

Sensitivity analysis — ranked, signed bar chart per parameter

`ui-2.js` Optimization page, `engine.js:446-463`

Design Optimization & Comparison

DONE

60-candidate cap, weighted multi-criteria scoring (Comfort 40% / Retention 25% / Solar 15% / Efficiency 10% / Cost 10%)

`engine.js:396, config.js:9-15` — weights match the checklist exactly

PARTIAL

Parameter sweep ranges

`engine.js:362-366`: orientation sweeps only 4 of 8 directions (South/SE/SW/East); insulation [50,75,100,150]mm (checklist also wants 25mm); window area [8,12,16,20]% (checklist also wants 10%); thermal mass [0,400,900,1600]kg (checklist wants [0,200,400,900]kg)

PARTIAL

Design comparison table

Top-ranked designs (A–E) are shown and compared; not confirmed to expose all 60 evaluated candidates in one sortable/filterable table, and no Excel export found

DONE

Recommended design highlighted, reasoning shown, “Adopt as current design” button

`ui-2.js:308 #adoptRecommendedBtn`

Reports & Documentation

DONE

Full engineering report: location, geometry, materials, thermal performance, heat flow, comfort score, recommendations, assumptions/limitations, metadata (timestamp, model version, simulation ID, data sources)

`ui-3.js` Reports page — verified against a live render this session

PARTIAL

Download button

“Print / Save as PDF” via the browser's print dialog, not a one-click generated-file download

MISSING

Standalone one-page Climate Profile PDF and Material Comparison Sheet (with Excel export)

Climate and material data are folded into the one main report; no separate printable artifacts or spreadsheet export found

Settings & Validation

DONE	Assumptions list (transient model, film coefficient, sky-longwave simplification, ground temp default, constant internal gains, PCM apparent-cp) ui-3.js:289-295
DONE	Honest validation status — no invented accuracy figure, no false standards-compliance claim, no fabricated date Consistently “Not yet validated” / “Field validation required before deployment” wherever validation status appears
MISSING	Units toggle (°C/°F, m/ft, W/Btu) ui-3.js:281-284 Settings shows units as fixed read-only labels, not a switchable control
MISSING	Dark mode toggle Single light theme only in styles.css, no toggle or dark palette
MISSING	Default location, default unit preferences, export preferences Not found in Settings
MISSING	API status indicator (connected/offline) No live connectivity indicator found

UI / UX & Navigation

PARTIAL	Guided 5-step workflow Steps are Location → Shelter → Materials → Comfort → Run; the checklist's step 4 is “Configure Openings/Windows” — in this app, windows/doors are only editable in the full Shelter Designer
DONE	Progress indicator, Next/Back navigation ui-1.js:520-536
DONE	Sidebar navigation with all listed sections (+ an extra Evaluator Summary page); Simple/Advanced mode now actually filters visible sections index.html, wired this session
DONE	Mobile-responsive hamburger sidebar styles.css:218-241, verified live this session
MISSING	Offline capability (cache 10 locations for no-network use) No service worker or bundled offline dataset — every location requires a live fetch on first load
MISSING	4-tier color-coded comfort zones as a visual design language (green/yellow/red/blue) Same gap as under Visualization — only a single comfort band color is used

Data Flow & Project Persistence

MISSING	Multiple named, savable/loadable projects store.js:31 — state holds exactly one project object (name + rename + reset); no project list or switcher
DONE	Session persistence across refresh, clear-project option localStorage-backed STORE; “Reset project (clears all data)” confirmed in Settings

Extra API Sources (checklist's own cross-validation wishlist)

The checklist itself frames these as enhancements beyond the two core sources — none are implemented, which is a reasonable state for a prototype rather than a Tier-1 gap.

MISSING

ERA5/Copernicus cross-check, IMD ground-station data, ISRO Bhuvan/SRTM elevation & terrain shading, Solcast, cross-source confidence badge

No references to any of these in the codebase; elevation remains a hardcoded reference table

Reliability, Crash Prevention & Caching

The checklist labels this section's items as Tier 1 (#15) — “a network hiccup crashing the live demo is a P0 risk, not polish.” None of it exists yet.

MISSING

Request timeouts, retry with backoff, circuit breaker on repeated API failures

No AbortController/timeout/retry logic anywhere in `weather-api.js` or `nasa-power.js` — a hung fetch has no time limit

PARTIAL

Caching

A flat 7-day TTL cache exists for both weather sources; not the short-TTL-live vs. long-TTL-normals split the checklist describes, and there's no DEM/elevation cache since no DEM is fetched

MISSING

Fallback chain (live → fresh cache → stale cache → bundled offline dataset)

Only live-or-cache-or-visible-error exists; deliberately no bundled offline climate data ships (a scientific-integrity choice per README), which also means zero fallback if a venue has no network

MISSING

Input validation before simulation (reject negative thickness, invalid lat/lon, zero ACH/volume)

Engine clamps some output aggregates to zero (`engine.js:266-270`) but doesn't reject bad inputs before they reach the solver

MISSING

Global exception handler, failure logging

Only local try/catch around specific async calls (e.g. `runLiveDemo`); no app-wide error boundary

Things the checklist says to remove — status

DONE

Hardcoded Ladakh demo dataset, placeholder material values, generic assumptions

Git history: “Remove illustrative Ladakh demo dataset entirely”; README states no hand-authored climate data ships with the app

PARTIAL

“Fixed Ladakh Demo button → replace with Load Real Weather”

The button is correctly relabeled “Run Live Demo” and fetches real live data (not fake), but `app.js:63` still hardcodes `loadRealClimate("leh")` — the one-click demo always targets Leh specifically, it just does so honestly now

Audited against the live codebase in `C:\AreaTherm\app\js\` during this session — every DONE/PARTIAL/MISSING above is backed by a `grep` or read of the actual source, not inferred from documentation.